Supersingular poles and signed residues of prime-level Fricke Hauptmoduln

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Abstract

Let N be one of the fifteen primes for which $X_0(N)^+$ has genus zero, let $T_N = q^{-1} + O(q)$ be its Hauptmodul, and let $p \geq 5$ be a prime with $p \neq N$. We determine the poles and signed residues of the rational function representing $T_N|U_p$ modulo p. Its poles are precisely the supersingular values with effective ramification index one, and each residue multiplied by the weighted product of differences from all other supersingular values equals $-\left(\frac{-N}{p}\right)$. Consequently the supersingular values and their indices determine the reduced function. The sign rests on Baker's supersingular congruence and on the signed discriminant-ratio identity of a companion paper. The indices also describe node thicknesses in the Deligne–Rapoport Fricke quotient. Exact computations provide independent finite certificates for the residue formula at all fifteen levels.

Keywords: supersingular elliptic curves; Fricke groups; Hauptmoduln; Hecke operators; eta quotients; congruences.

1 Introduction

This paper concerns the reduction of U_p applied to a Fricke Hauptmodul. At each of the fifteen prime levels with genus-zero Fricke quotient, we determine the pole divisor and signed residues of the reduced function. The ramification indices of the quotient remove certain supersingular points from the pole divisor, but those points remain in the weighted residue product. The formula reconstructs the entire reduced function from the supersingular values and their indices.

The sign in the residue formula comes from two supersingular identities, recalled in Theorem 1.1: Baker's congruence (Baker, 1998) and the signed discriminant-ratio identity proved in the companion paper (Jordan, 2026).

1.1 Notation

Let $p \geq 5$ be prime and put

$$h = (p-1)/2, \qquad k = (p^2-1)/24, \qquad A = \overline{E}_{p-1}, \qquad B = \overline{E}_{p+1}, \qquad \Delta = \eta^{24}.$$

Here E_w is the normalized Eisenstein series of weight w, a bar denotes reduction modulo p, and A is the Hasse invariant. The q-expansions of A and B are 1 and E_2 modulo p, respectively. We write $q = e^{2\pi i\tau}$, $\theta = q\,d/dq$, $\mathbb{F}_p = \mathbb{F}_p$, $\mathbb{F}_{p^2} = \mathbb{F}_{p^2}$ and $\overline{\mathbb{F}}_p = \overline{\mathbb{F}}_p$. For $f = \sum c_n q^n$, define $f|U_p = \sum c_{pn} q^n$. The symbol $\left(\frac{a}{p}\right)$ denotes the Legendre symbol. When a modular form is evaluated at an elliptic curve, a nonzero invariant differential is understood. Ratios of equal weight are independent of its choice. In particular, set

$$\beta(E) := \frac{B(E,\omega)^{p-1}}{\Delta(E,\omega)^{2k}}.$$

1.2 Two supersingular identities

For a prime $N \neq p$, put $D_N = \Delta(\tau)/\Delta(N\tau)$. It is a ratio of two forms of weight 12 on $\Gamma_0(N)$ with integral q-expansions, neither of which vanishes on the non-cuspidal curve $Y_0(N)$ over $\mathbb{Z}[1/N]$. It is therefore a unit on $Y_0(N)$, and we use the same notation for its reduction modulo p. A description of D_N in terms of the isogeny $E \to E/C$ is given in Jordan (2026).

Theorem 1.1 (supersingular inputs)**.** *Let $p \geq 5$ and $N \neq p$ be primes.*

(i) (Baker, 1998, Theorem 1.4). *For every supersingular elliptic curve $E/\overline{\mathbb{F}}_p$,*

$$\beta(E) = -\left(\frac{-1}{p}\right).$$

(ii) (Jordan, 2026, Theorem 1.3). *At every supersingular point x of $Y_0(N)_{\overline{\mathbb{F}}_p}$,*

$$D_N(x)^{(p^2-1)/24} = \left(\frac{N}{p}\right).$$

Part (ii) refines the identity $D_N(x)^{(p^2-1)/12} = 1$ of Ono (2026, Theorem 1.2). It is deduced in Jordan (2026) from the evaluation $B(E,\omega)^h = \left(\frac{6}{p}\right) h!\,\Delta(E,\omega)^k$ on the supersingular locus, which also recovers part (i).

1.3 The residue formula

Throughout,

$$N \in \mathcal{N} := \{2,3,5,7,11,13,17,19,23,29,31,41,47,59,71\},$$

and $p \geq 5$ is prime with $p \neq N$. These are precisely the prime levels for which $X_0(N)^+$ has genus zero (Ogg, 1975, §3, Corollary). Let T_N denote its unique Hauptmodul with expansion $q^{-1} + O(q)$ at the single cusp. Lemma 2.1 constructs this coordinate over $\Lambda_N := \mathbb{Z}[1/(6N)]$ and proves that it remains a coordinate at every admissible prime. The standard moonshine identification relates these functions to prime-order Fricke McKay–Thompson series (Chen et al., 2019; Conway and Norton, 1979); the proof below uses the modular curves.

For the explicit applications at $N \in \mathcal{N}_0 := \{2,3,5,7,13\}$ we also use

$$a_N = \frac{24}{N-1}, \qquad b_N = N^{12/(N-1)}, \qquad u_N = \left(\frac{\eta(\tau)}{\eta(N\tau)}\right)^{a_N}, \qquad T_N = u_N + a_N + b_N/u_N.$$

The pairs (a_N, b_N) are $(24, 4096), (12, 729), (6, 125), (4, 49), (2, 13)$. Here u_N is a coordinate on $X_0(N)$ and w_N sends it to b_N/u_N. This eta presentation is not used to define T_N at the ten other levels.

Write $T = T_N$ and $\Delta_N = \Delta(\tau)\Delta(N\tau)$. The level N and the reduction prime p play distinct roles: p need not belong to $\mathcal{N}$, and $p = N$ is excluded. At the five eta-coordinate levels, $D_N = u_N^{N-1}$, so Theorem 1.1(ii) gives

$$u_N(x)^{(p^2-1)/a_N} = \left(\frac{N}{p}\right)$$

at every supersingular point x of $Y_0(N)$.

The following weights distinguish the supersingular points that contribute poles from those at which ramification cancels a pole.

Let $t_1, \ldots, t_s \in \overline{\mathbb{F}}_p$ be the distinct supersingular values of T. For a point $x = (E, C)$ of $X_0(N)_{\overline{\mathbb{F}}_p}$ over t_i put

$$m_x = |\mathrm{Aut}(E,C)/\{\pm 1\}|, \qquad e_i = m_x\,|\mathrm{Stab}_{\langle w_N\rangle}(x)|.$$

Call t_i *thin* if $e_i = 1$ and *thick* if $e_i > 1$; the terms refer to the node-thickness interpretation of Section 5. Put $D_{\mathrm{thin}}(T) = \prod_{e_i=1}(T - t_i)$. With $R_{N,p}$ the rational function of Theorem 1.2(a) below, let, for thin t_i,

$$r_i = \mathrm{Res}_{T=t_i} R_{N,p}(T), \qquad W_i = \prod_{j \neq i} (t_i - t_j)^{(p+1)/e_j}.$$

Section 3 shows that e_i does not depend on the representative of the Fricke orbit, that it is the effective (tame) inertia of the quotient map at x (not the order of a group containing ± 1), and that every exponent $(p+1)/e_j$ is an integer.

Only thin values give poles of $R_{N,p}$, and every such pole is simple (Theorem 1.2(b)).

Theorem 1.2 (Fricke residue formula)**.** *Let $N \in \mathcal{N}$ and let $p \geq 5$ be a prime with $p \neq N$.*

(a) ***(Rationality.)*** *There is a unique $R_{N,p} \in \mathbb{F}_p(T)$ with $\overline{T|U_p} = R_{N,p}(\overline{T})$. It vanishes at $T = \infty$.*

(b) ***(Pole support.)*** *$R_{N,p} = P/D_{\mathrm{thin}}$ with $P \in \mathbb{F}_p[T]$, $\deg P < \deg D_{\mathrm{thin}}$ and $\gcd(P, D_{\mathrm{thin}}) = 1$. Every pole is simple and has a nonzero residue. In particular $\mathrm{den}_{\mathrm{red}} R_{N,p} = D_{\mathrm{thin}}$.*

(c) ***(Residue as a Hasse derivative.)*** *If t_i is thin and x lies over t_i, then*

$$r_i = \frac{144\,(\theta T)(x)^{p+1}}{B(x)^2}, \qquad \theta = q\frac{d}{dq},$$

a ratio of two mod-p modular forms of weight $2(p+1)$, evaluated at x.

(d) ***(Product.)*** *For thin t_i and $x = (E, C)$ over t_i,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \beta(E)\, D_N(x)^k.$$

(e) ***(Signed weighted residue formula.)*** *For every thin t_i,*

$$\boxed{r_i W_i = -\left(\frac{-N}{p}\right)} \qquad \text{equivalently} \qquad r_i = -\left(\frac{-N}{p}\right) \prod_{j \neq i} (t_i - t_j)^{-(p+1)/e_j}.$$

In particular $r_i \in \mathbb{F}_p^\times$ and $W_i \in \mathbb{F}_p^\times$, also when $t_i \notin \mathbb{F}_p$.

(f) ***(Hecke form.)*** *With $\Psi_{N,p}(X,Y) = (X^p - Y)(X - Y^p) + pH_{N,p}(X,Y)$ as in Proposition 2.3, $\overline{H}_{N,p}(t^p, t) = \left(\frac{-N}{p}\right) W_i^{-1}$ if $t = t_i$ is thin, and $\overline{H}_{N,p}(t^p, t) = 0$ for every other $t \in \mathbb{F}_{p^2}$.*

Corollary 1.3 (reconstruction)**.** *The supersingular values and their indices determine the entire reduced function:*

$$R_{N,p}(T) = -\left(\frac{-N}{p}\right) \sum_{e_i = 1} \frac{1}{W_i (T - t_i)}. \tag{1.1}$$

Equivalently, if $D_{\mathrm{thin}} = \prod_f f$ is its factorization into monic irreducibles over $\mathbb{F}_p$, then

$$R_{N,p}(T) = \sum_{f \mid D_{\mathrm{thin}}} c_f \frac{f'(T)}{f(T)}, \qquad c_f = -\left(\frac{-N}{p}\right) W_i^{-1} \in \mathbb{F}_p^\times \quad \text{for any root } t_i \text{ of } f. \tag{1.2}$$

Both sums are empty when $D_{\mathrm{thin}} = 1$.

Proof. Subtracting the principal parts leaves a polynomial, which is zero because the function and each summand vanish at infinity. Frobenius-conjugate poles have conjugate residues; the residues lie in $\mathbb{F}_p$ by Theorem 1.2(e), so they are equal on each irreducible factor. The identity $f'/f = \sum_{f(t_i)=0} (T - t_i)^{-1}$ gives the second formula. □

The residues determine the reduced function because it has only simple poles and vanishes at the cusp. Corollary 1.3 therefore reconstructs $R_{N,p}$ from the supersingular values and their indices alone. Remark 6.2 compares the size of these data with that of a q-expansion certificate. The same effective indices are interpreted as node thicknesses in Section 5.

1.4 Consequences of the Fricke application

The reconstruction in Corollary 1.3 gives concrete formulas once the supersingular values and their effective indices are known. At the five levels $N \in \mathcal{N}_0$, the connection to Nakaya's squarefree Fricke supersingular polynomials is a change of coordinate: with the normalizations in Nakaya (2019, §4),

$$\mathcal{S}_{N,p}(T) := \prod_i (T - t_i) = ss_p^{(N*)}(T + c_N), \qquad \begin{array}{c|ccccc} N & 2 & 3 & 5 & 7 & 13 \\ \hline c_N & 104 & 42 & 16 & 9 & 0 \end{array} \tag{1.3}$$

For $N = 2, 3, 5, 7$, Nakaya's printed eta formulas have additive constants $128, 54, 22, 13$, giving the shifts displayed above. At level 13 his convention is zero constant term in the q-expansion; this identifies the normalized Hauptmodul with T_{13} and gives $c_{13} = 0$. All constants are reduced modulo p; $\mathcal{S}_{N,p}$ is distinct from the weighted polynomial S of Proposition 3.4. Thus $D_{\text{thin}} = \mathcal{S}_{N,p} / \prod_{e_i > 1}(T - t_i)$, while the omitted values still occur in the weights W_i. For these five levels the exceptional values can also be specified by a polynomial independent of p. Let $\mathfrak{h}_N$ be the elliptic-value polynomial in Table 2. Lemma 3.2 and Theorem 1.2(b) give

$$D_{\text{thin}}(T) = \frac{\mathcal{S}_{N,p}(T)}{\gcd\big(\mathcal{S}_{N,p}(T), \overline{\mathfrak{h}}_N(T)\big)}, \tag{1.4}$$

where the gcd is monic. Thus the denominator is obtained from Nakaya's supersingular polynomial by removing its common factors with the reduction of a fixed characteristic-zero elliptic-value polynomial.

At level two, put $\varepsilon_p = (1 - \left(\frac{-1}{p}\right))/2$ and $\nu_p = (1 - \left(\frac{-2}{p}\right))/2$. The level-two indices described in Section 5 give

$$D_{2,p}(T) = \frac{ss_p^{(2*)}(T + 104)}{(T + 104)^{\varepsilon_p}(T - 152)^{\nu_p}}. \tag{1.5}$$

The two removed values correspond to 0 and 256 in Nakaya's coordinate; they are distinct for $p \geq 5$.

Corollary 1.4 (level-two degree and vanishing)**.** *For every prime $p \geq 5$,*

$$\deg D_{2,p} = \frac{p - 4 + \left(\frac{-1}{p}\right) + 2\left(\frac{-2}{p}\right)}{8}. \tag{1.6}$$

In particular $T_{2A}|U_p \equiv 0 \pmod{p}$ if and only if $p = 5$ or 7.

Proof. The two possible thick values have indices four and two, and supersingularity indicators ε_p and ν_p, respectively. Corollary 3.5 therefore gives $\deg D_{2,p} = (p-1)/8 - \varepsilon_p/4 - \nu_p/2$, which is (1.6). This degree is zero at $p = 5, 7$ and is at least $(p-7)/8 > 0$ for $p \geq 11$. Theorem 1.2(b) proves the last assertion. □

Chen et al. (2019, Lemma 4.8 and Table 4.3) establish p-adic annihilation of T_{2A} at $p = 5, 7$. The corollary specifies the exact first-step modulo-p vanishing criterion. Table 1 illustrates the nonzero cases; at $p = 17$, for example, $R_{2,17} = D'_{2,17}/D_{2,17}$ and both residues are one.

1.5 Relation to earlier work

The sign input, Theorem 1.1, combines Baker (1998) with Jordan (2026); the remaining ingredients are as follows.

The derivative argument in Section 4 follows the level-one method presented by Kaneko and Zagier (1998, §10), who trace it to Koike's work on Atkin's congruences (Koike, 1973). The period formula of de Shalit (1994) and Baker's congruence supply the level-one precedents for

p	$D_{2,p}(T)$	$P_{2,p}(T)$
11	$T-5$	1
13	$T-3$	-2
17	T^2-T-8	$2T-1$
29	$T^3-12T^2+12T-6$	$4T^2-5T-8$

Table 1: Level-two examples, with $R_{2,p}=P_{2,p}/D_{2,p}$. Coefficients use the unique integer representatives in $(-p/2,p/2)$. These entries are included in the certificates of Section 6.

the residue product and its sign; Betina and Lecouturier (2018) treat the corresponding Legendre modular polynomials. The Fricke application combines this derivative method with Theorem 1.1 and the effective ramification indices of the quotient.

Fricke supersingular polynomials are studied by Sakai (2011, 2014), Nakaya (2019) and Morton (2023). These papers treat orthogonal-polynomial representations, factorization and resultant identities. Formula (1.3) identifies the supersingular values used here with that literature; the present residue formula additionally assigns a nonzero coefficient to each thin pole and retains the ramified values in its weighted product.

Jorgenson et al. (2019, §4.2, Table 1 and Appendix C) determine the elliptic values of normalized Fricke Hauptmoduln and their associated class fields. Their j_N equals our T_N. Table 2 records their polynomials for all fifteen prime levels. Their Schwarzian equation detects the same local ramification indices that enter our Hasse-derivative calculation; (1.4) expresses its elliptic-value locus as the locus of possible supersingular pole cancellations. The signed residue formula concerns the remaining thin values.

Duncan and Swisher (2026, Proposition 3.1 and §4) use Deligne's supersingular expansion of $pJ|U_p$ and prime-power level modular functions to characterize the monstrous exponents. Here we determine the signed residues on the fifteen prime-level genus-zero Fricke curves.

Bringmann et al. (2026, Theorem 1.1) study iterated U_p congruences for positive-weight meromorphic cusp forms on $\mathrm{SL}_2(\mathbb{Z})$ whose interior pole reduces to a supersingular point. Their p-adic approximation by holomorphic cusp forms concerns a different setting from our first-step modulo-p pole divisor and signed residue formula.

Organization. Section 2 constructs the integral coordinate and proves the residue criterion. Section 3 treats the effective indices and the weighted supersingular polynomial. Section 4 proves Theorem 1.2. Section 5 interprets the weights geometrically, and Section 6 records finite certificates and exact-arithmetic evidence.

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2 Standard inputs and the residue criterion

Lemma 2.1 (integral Fricke coordinate)**.** *Let N be prime with $g(X_0(N)^+)=0$, and put $\Lambda_N=\mathbb{Z}[1/(6N)]$. The coarse quotient*

$$\mathcal{X}_N^+ := X_0(N)_{\Lambda_N}/\langle w_N\rangle$$

is a smooth proper curve over Λ_N with a single cusp section c. There is a unique coordinate T_N over Λ_N whose expansion at that section is $q^{-1}+O(q)$. It identifies $\mathcal{X}_N^+$ with $\mathbb{P}^1_{\Lambda_N}$ and c

with infinity. In particular, $T_N \in q^{-1} + q\Lambda_N[[q]]$ and its reduction is a coordinate at every prime $p \geq 5$ with $p \neq N$.

Proof. Away from $6N$ the coarse modular curve $X_0(N)$ is smooth and proper, with geometrically connected fibres; its two disjoint rational cusp sections are exchanged by w_N (Deligne and Rapoport, 1973; Katz and Mazur, 1985). The quotient exists and is proper over Λ_N. To see that it is flat, work on an invariant affine open with coordinate ring A. Since 2 is invertible in Λ_N, the averaging map

$$a \longmapsto \frac{a + w_N(a)}{2}$$

makes the invariant ring $A^{\langle w_N \rangle}$ a direct summand of A. As A is flat over Λ_N, its direct summand is also flat. Since 2 is invertible, formation of the quotient commutes with base change. Each geometric fibre is the quotient of a smooth curve by an involution, hence is smooth; thus $\mathcal{X}_N^+$ is smooth over Λ_N. Its fibres are geometrically connected and have constant genus, which is zero by the assumption on the generic fibre. The two cusps give one section c. The quotient map is an isomorphism on the completed neighbourhood of either cusp onto that of c, so $\widehat{\mathcal{O}}_{\mathcal{X}_N^+,c} = \Lambda_N[[q]]$.

Write $f : \mathcal{X}_N^+ \to \operatorname{Spec} \Lambda_N$. On every fibre, $\mathcal{O}(c)$ has degree one and $H^1(\mathcal{O}) = 0$. Cohomology and base change therefore make $f_*\mathcal{O}(c)$ locally free of rank two, and the principal-part sequence is exact:

$$0 \longrightarrow \Lambda_N \longrightarrow H^0(\mathcal{X}_N^+, \mathcal{O}(c)) \longrightarrow \Lambda_N\, q^{-1} \longrightarrow 0.$$

The last identification uses the parameter q to trivialize the normal line of the section. Choose a section with principal part q^{-1} and subtract its constant coefficient. This gives T_N; two such choices differ by a global constant, proving uniqueness. The sections $1, T_N$ form a basis on every fibre and generate $\mathcal{O}(c)$ there, so they give an isomorphism to $\mathbb{P}^1_{\Lambda_N}$, with the asserted cusp and coordinate. □

The lemma uses genus zero only for the Fricke quotient. Neither its proof nor the arguments below require $X_0(N)$ itself to have genus zero.

We use the following standard results for prime-to-p level structures. The q-expansion principle is Katz (1973, §1.6). The Hasse invariant, its q-expansion and Deligne's congruence $A \equiv E_{p-1} \pmod{p}$ are treated in Katz (1973, §§2.0–2.1). That the Hasse invariant has simple zeros, precisely at the supersingular points of a fine moduli cover, is Igusa's theorem (Katz and Mazur, 1985, Ch. 12). The Kodaira–Spencer isomorphism, under which a weight-two form f corresponds to $f\,dq/q$, is in Katz (1973, Appendix 1); see also Gross (1990). In particular, θT represents dT. We record the statements used repeatedly.

Standard fact I1.

(i) The q-expansion principle holds at prime-to-p level. The forms A and B have q-expansions 1 and E_2 modulo p, respectively.

(ii) The Hasse invariant A has simple zeros precisely at the supersingular points on a fine moduli cover.

(iii) For $p \geq 5$ the stabilizers relevant to $X_0(N)$ and its Fricke quotient are tame, and these coarse quotients commute with reduction. The supersingular values of T lie in $\mathbb{F}_{p^2}$.

(iv) Kodaira–Spencer identifies a weight-two form f with $f\,dq/q$; in particular θT represents dT.

The field-of-definition assertion in (iii) follows by choosing a model with p^2-Frobenius $[-p]$ as follows; every cyclic subgroup of order N is then rational as a subgroup scheme.

Lemma 2.2 (Deuring models)**.** *Let $E_1/\overline{\mathbb{F}}_p$ be supersingular, $p \geq 5$. Then E_1 has a model over $\mathbb{F}_{p^2}$ whose p^2-power Frobenius endomorphism is $[-p]$. For such a model every finite subgroup of order prime to p is $\mathbb{F}_{p^2}$-rational, and E_1/K again has Frobenius $[-p]$.*

Proof. A supersingular curve admits a model over $\mathbb{F}_{p^2}$ (Deuring, 1941; Waterhouse, 1969). For any such model $E/\mathbb{F}_{p^2}$, the p-power Frobenius $\mathrm{Fr} : E \to E^{(p)}$ and the Verschiebung $V : E^{(p)} \to E$ are both purely inseparable of degree p (the Hasse invariant vanishes), so $\mathrm{Fr}^{(p)} : E^{(p)} \to E^{(p^2)} = E$ and V differ by an automorphism u of E: $\mathrm{Fr}^{(p)} = u \circ V$, whence $\pi_{p^2} = \mathrm{Fr}^{(p)} \circ \mathrm{Fr} = u \circ V \circ \mathrm{Fr} = u \circ [p]$. Thus $\pi_{p^2} = p\,u$ with $u \in \mathrm{Aut}(E)$. If $j \neq 0, 1728$ then $u = \pm 1$ and the quadratic twist changes the sign; if $j = 0$ (resp. 1728) the curve $y^2 = x^3 + 1$ (resp. $y^2 = x^3 + x$) over $\mathbb{F}_p$ has trace 0, so $\pi_{p^2} = \pi_p^2 = [-p]$ (Deuring, 1941; Waterhouse, 1969); see also Ono (2026, Lemma 4.3). Since $[-p]$ is central, Galois preserves every subgroup, and the quotient isogeny ϕ satisfies $\pi'\phi = \phi\pi = \phi[-p] = [-p]\phi$, so $\pi' = [-p]$. □

Proposition 2.3. *Let Δ_p be the set of integral matrices of determinant p with lower-left entry divisible by N. There is a unique $\Psi_{N,p} \in \Lambda_N[X, Y]$ with*

$$\Psi_{N,p}(X, T(\tau)) = (X - T(p\tau)) \prod_{b=0}^{p-1} \Big(X - T\big(\tfrac{\tau+b}{p}\big)\Big).$$

It is monic of degree $p+1$ in each variable, symmetric, and satisfies $\Psi_{N,p} \equiv (X^p - Y)(X - Y^p) \pmod{p}$.

Proof. The matrices $\mathrm{diag}(p, 1)$ and $\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$, $0 \leq b < p$, represent $\Gamma_0(N)\backslash\Delta_p$. Conjugation by $W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ maps $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to $\begin{pmatrix} d & -c/N \\ -Nb & a \end{pmatrix}$, so it preserves Δ_p. Hence the elementary symmetric functions of the roots are $\Gamma_0(N)^+$-invariant, holomorphic on the upper half-plane, with a pole only at the single cusp. Their coefficients initially lie in $\Lambda_N[\zeta_p]((q^{1/p}))$. Translation of τ removes fractional powers, and cyclotomic Galois automorphisms permute the roots, so the expansions lie in $\Lambda_N((q))$. Subtracting powers of $T = q^{-1} + O(q)$ then puts them in $\Lambda_N[T]$. Modulo $1 - \zeta_p$ all $T((\tau + b)/p)$ coincide with $T(q^{1/p})$, whose p-th power is $T(q)$, and $T(p\tau) \equiv T^p$; this gives the congruence. For irreducibility, the reduction of $\Gamma_0(N)$ modulo p is $\mathrm{SL}_2(\mathbb{F}_p)$, by the Chinese remainder theorem and the surjectivity of reduction for $\mathrm{SL}_2(\mathbb{Z})$. It acts transitively on the $p+1$ cyclic subgroups of order p. The $p+1$ roots displayed above are distinct: their leading terms are q^{-p} and $\zeta_p^{-b} q^{-1/p}$ for $0 \leq b < p$. They therefore form one orbit, so $\Psi_{N,p}(X, T)$ is irreducible over $\mathbb{C}(T)$. The pole order of the jth elementary symmetric function is at most $p + (j-1)/p \leq p + 1$; the constant coefficient has pole order $p + 1$. Thus the polynomial has degree $p + 1$ in Y as well as in X. For symmetry, $\mathrm{diag}(1, p) \in \Delta_p$ gives $\Psi_{N,p}(T(\tau), T(p\tau)) = 0$ by evaluating the defining relation at $p\tau$. Irreducibility identifies the reversed polynomial with the original up to a rational scalar. The constant coefficient, as a polynomial in X, has leading term $q^{-(p+1)}$ with coefficient one, so $\Psi_{N,p}$ is monic of degree $p + 1$ in Y. Comparing the X^{p+1} coefficients in the two polynomials makes that scalar one. Thus $\Psi_{N,p}(Y, X) = \Psi_{N,p}(X, Y)$. □

Define $H = H_{N,p} \in \Lambda_N[X, Y]$ by $\Psi_{N,p} = (X^p - Y)(X - Y^p) + pH$; H is symmetric. Put

$$\psi = \frac{T(q)^p - T(q^p)}{p} \in \Lambda_N((q)),$$

which lies in this ring because $T(q^p) \equiv T(q)^p \pmod{p\Lambda_N}$.

Proposition 2.4 (residue criterion)**.** *In $\mathbb{F}_p((q))$,*

$$\overline{\psi} = \frac{\overline{H}(T^p, T)}{T^{p^2} - T}, \qquad \overline{T|U_p} = \overline{Q}(T) + \overline{\psi},$$

where $\overline{Q} \in \mathbb{F}_p[T]$ *is defined by* $-[X^p]\Psi_{N,p} = Y^p + pQ(Y)$. *Hence* $R_{N,p} = \overline{Q}(T) + \overline{H}(T^p, T)/(T^{p^2} - T)$ *exists and is unique. It vanishes at* ∞, *its poles are simple and lie in* $\mathbb{F}_{p^2}$, *and for* $t \in \mathbb{F}_{p^2}$

$$\operatorname{Res}_{T=t} R_{N,p} = -\overline{H}(t^p, t).$$

Proof. Substitute $X = T(q^p) = T^p - p\psi$ and $Y = T$ into $\Psi_{N,p} = 0$. Since $X - Y^p = -p\psi$, division by p and reduction give $-\overline{\psi}\,(T^{p^2} - T) + \overline{H}(T^p, T) = 0$. The sum of the roots of $\Psi_{N,p}(X, T)$ is $T(q^p) + p\,(T|U_p) = T^p + pQ(T)$, so $T|U_p = Q(T) + \psi$. The polynomial $T^{p^2} - T$ is squarefree, its roots are exactly $\mathbb{F}_{p^2}$, and its derivative is -1. $R_{N,p}$ vanishes at ∞ because $T|U_p \in q\Lambda_N[[q]]$ (the constant term of T is 0). Uniqueness holds because $\overline{T}$ is transcendental over $\mathbb{F}_p$. □

Remark 2.5 (Faber polynomials and replication). The polynomial

$$\mathcal{F}_{N,p}(X) := -[Z^p]\Psi_{N,p}(Z, X) = X^p + pQ(X)$$

is the degree-p Faber polynomial of T: it is the unique monic polynomial for which $\mathcal{F}_{N,p}(T(q)) = q^{-p} + O(q)$. Indeed, the sum-of-roots identity used above gives

$$\mathcal{F}_{N,p}(T(q)) = T(q^p) + p(T|U_p)(q).$$

Under the standard McKay–Thompson identification of T, this is the prime-to-N replication identity; see Conway and Norton (1979, §8) for the replication terminology. The polynomial Q is the correction $(\mathcal{F}_{N,p} - X^p)/p$, rather than the Faber polynomial itself. Proposition 2.4 describes its reduction together with the first Kronecker correction.

Corollary 2.6. *For every* $t \in \mathbb{F}_{p^2}$, $\operatorname{Res}_t R_{N,p} \in \mathbb{F}_p$ *and* $\operatorname{Res}_{t^p} R_{N,p} = \operatorname{Res}_t R_{N,p}$.

Proof. $(\operatorname{Res}_t R)^p = \operatorname{Res}_{t^p} R = -\overline{H}(t^{p^2}, t^p) = -\overline{H}(t, t^p) = -\overline{H}(t^p, t) = \operatorname{Res}_t R$, using $t^{p^2} = t$, the $\mathbb{F}_p$-rationality of R and the symmetry of H. □

Proposition 2.4 gives Theorem 1.2(a) and the a priori bound $\operatorname{den}_{\mathrm{red}} R_{N,p} \mid T^{p^2} - T$ used by the certificates in Section 6.

3 Weights and the weighted supersingular polynomial

Lemma 3.1 (effective inertia)**.** *Let* $x = (E, C)$ *be a non-cuspidal point of* $X_0(N)_{\overline{\mathbb{F}}_p}$, *supersingular or ordinary. Let* G_x *be the group of automorphisms of* (E, C) *together with the isomorphisms* $(E, C) \to w_N(E, C) = (E/C, E[N]/C)$ *when* $w_N(x) = x$. *It is a group once* $w_N^2(E, C) \cong (E, C)$ *is identified via* $[N]$.

(i) $|G_x/\{\pm 1\}| = m_x |\operatorname{Stab}_{\langle w_N \rangle}(x)| = e_x$. *The value is the same at* x *and at* $w_N(x)$.

(ii) $\{\pm 1\}$ *is exactly the kernel of the action of* G_x *on the deformation space of* x. *Since* $e_x \in \{1, 2, 3, 4, 6\}$ *is prime to* p, $G_x/\{\pm 1\}$ *is cyclic and acts faithfully on the tangent line. Hence, for a local parameter* z *of a fine moduli cover at a point over* x, $T - t = z^{e_x} \cdot (\text{unit})$.

(iii) *If* x *is supersingular,* $(p + 1)/e_x \in \mathbb{Z}$.

Proof. (i) If $w_N(x) = x$, the isomorphisms $(E, C) \to w_N(E, C)$ form a torsor under $\operatorname{Aut}(E, C)$, so $|G_x| = 2|\operatorname{Aut}(E, C)|$. Conjugation by the isogeny $E \to E/C$ identifies $\operatorname{Aut}(E, C)$ with $\operatorname{Aut}(E/C, E[N]/C)$, so $m_x = m_{w_N(x)}$. (ii) $[-1]$ acts trivially on deformations. A Fricke-type element acts on the deformation functor by $(\widetilde{E}, \widetilde{C}) \mapsto (\widetilde{E}/\widetilde{C}, \widetilde{E}[N]/\widetilde{C})$, followed by the chosen identification of $w_N(E, C)$ with (E, C). If its action were trivial, the universal deformation would admit an isomorphism $\widetilde{E}/\widetilde{C} \cong \widetilde{E}$. Composing with the quotient map would give a degree-N endomorphism of its generic fibre, whose endomorphism ring is $\mathbb{Z}$; this is impossible since N is

not a square. Similarly, a nontrivial automorphism other than ± 1 acting trivially would extend to the universal deformation and contradict that generic endomorphism ring. The same argument works at ordinary points. A finite group of order prime to p acting faithfully on $k[[z]]$ is cyclic and acts faithfully on $(z)/(z^2)$, so the invariant ring is $k[[z^e]]$. (iii) $e = 3$ or 6 forces $m_x = 3$, so $j = 0$ is supersingular and $p \equiv 2 \pmod 3$; $e = 4$ forces $m_x = 2$, so $j = 1728$ is supersingular and $p \equiv 3 \pmod 4$. For $p \geq 5$, $m_x \in \{1, 2, 3\}$. □

For explicit calculations at $N \in \mathcal{N}_0$, write $u = u_N$. Then m_x can be read off from the multiplicity μ of $u(x)$ as a root of $P_N(u) - j\, u^N$: $m_x = 3/\mu$ over $j = 0$, $m_x = 2/\mu$ over $j = 1728$, and $m_x = 1$ (with $\mu = 1$) otherwise. Here μ is the ramification index of the coarse map $X_0(N) \to X(1)$. The Fricke stabilizer has order two exactly when $u(x)^2 = b_N$.

Lemma 3.2 (elliptic values and specialization)**.** *Let $N \in \mathcal{N}$ and let $\mathfrak{h}_N(T) = \prod_\tau (T - T_N(\tau))$, where τ runs over the inequivalent elliptic points of $\Gamma_0(N)^+$. The fifteen polynomials and their effective indices are given in Table 2. For every prime $p \geq 5$ with $p \neq N$, the reduction $\overline{\mathfrak{h}}_N$ is squarefree, and its roots in $\overline{\mathbb{F}}_p$ are exactly the values with effective index $e > 1$. Consequently,*

$$\gcd(\mathcal{S}_{N,p}, \overline{\mathfrak{h}}_N) = \prod_{e_i > 1} (T - t_i).$$

Proof. The characteristic-zero polynomials are those of Jorgenson et al. (2019, §4.2, Table 1), with $T_N = q^{-1} + O(q)$. Their elliptic orders give the effective indices of Lemma 3.1. For $N \in \mathcal{N}_0$, the Fricke fixed values are given by $u_N^2 = b_N$. At the remaining levels, the smooth models $v^2 = f_N(u)$ of Section 6 have involution $(u, v) \mapsto (u, -v)$ and coordinate $T_N = u - s_N$; the branch values are the roots of $f_N(T + s_N)$, all of index two. The additional factors are $T + 2$ of index two at $N = 17, 29$, T of index two at $N = 41$, $T + 3$ of index three at $N = 19$, and T of index three at $N = 31$. These are obtained from the elliptic fibres over $j = 1728$ and $j = 0$.

We justify specialization of the support. An effective stabilizer arises from an automorphism of (E, C) or a Fricke fixed point. For $p \geq 5$, nontrivial reduced automorphisms occur only at $j = 0, 1728$; their prime-to-p automorphisms lift with the corresponding CM elliptic curve, and an invariant prime-to-p cyclic subgroup lifts uniquely over a strict henselian base. Conversely, these automorphisms persist on reduction. The fixed-point scheme of w_N on $X_0(N)_{\Lambda_N}$ is finite étale over Λ_N. To see this, let x be a fixed geometric point of a fibre. After passing to the strict henselization of the base and completing, the local ring at x is $\mathcal{O}[[z]]$, where $\mathcal{O}$ is a complete local ring in which 2 is invertible. The involution acts on the cotangent line of the fibre at x by a sign. The sign is -1: a tame automorphism of finite order of a smooth connected curve that fixes a point and acts trivially on its cotangent line is the identity, whereas w_N exchanges the two cusps. Hence $z' = (z - w_N(z))/2$ is again a parameter, $\mathcal{O}[[z]] = \mathcal{O}[[z']]$, and $w_N(z') = -z'$. The ideal generated by all $w_N(f) - f$ is then $(2z') = (z')$, so the fixed-point scheme is $\operatorname{Spec} \mathcal{O}$ near x, a section. Being also proper and quasi-finite, the fixed-point scheme is finite étale. Fricke fixed points therefore neither appear nor merge under reduction at a prime not dividing $2N$. Thus the listed elliptic values specialize to precisely the values with $e > 1$.

Exact discriminant calculation for every displayed $\mathfrak{h}_N$ gives

$$\operatorname{Disc}(\mathfrak{h}_N) = \pm 2^{\lambda_N} 3^{\mu_N} N^{\nu_N} \qquad (\lambda_N, \mu_N, \nu_N \geq 0).$$

The exact values and model calculations are recorded in the supplementary material (Section 6). Hence distinct elliptic values cannot collide at any admissible prime, so $\overline{\mathfrak{h}}_N$ is squarefree. Intersecting its root set with the supersingular locus proves the gcd formula. □

Lemma 3.3. *As mod-p forms on $\Gamma_0(N)$: $A|_{p-1} W_N = \left(\frac{N}{p}\right) A$, $\Delta_N|_{24} W_N = \Delta_N$ and $\theta T|_2 W_N = \theta T$.*

N	$\mathfrak{h}_N(T)$	Effective indices
2	$(T+104)(T-152)$	4, 2
3	$(T+42)(T-66)$	6, 2
5	$(T+16)(T^2-12T-464)$	2, 2
7	$(T+10)(T+9)(T-18)$	2, 3, 2
11	$(T+6)(T^3-2T^2-76T-212)$	2, 2
13	$(T+4)(T+3)(T^2-4T-48)$	2, 3, 2
17	$(T+2)(T^4+2T^3-39T^2-176T-212)$	2, 2
19	$(T+4)(T+3)(T^3-4T^2-16T-12)$	2, 3, 2
23	$(T^3-2T^2-17T-25)$ $\cdot(T^3+6T^2+11T+7)$	2, 2
29	$(T+2)(T^6+2T^5-17T^4-66T^3-83T^2-32T-4)$	2, 2
31	$T(T^3-17T-27)(T^3+4T^2+3T+1)$	3, 2, 2
41	$T\big(T^8+4T^7-8T^6-66T^5-120T^4$ $-56T^3+53T^2+36T-16\big)$	2, 2
47	$(T^5-5T^3-20T^2-24T-19)$ $\cdot(T^5+4T^4+7T^3+8T^2+4T+1)$	2, 2
59	(T^3+2T^2+1) $\cdot(T^9+2T^8-4T^7-21T^6-44T^5-60T^4-61T^3-46T^2-24T-11)$	2, 2
71	$(T^7-7T^5-11T^4+5T^3+18T^2+4T-11)$ $\cdot(T^7+4T^6+5T^5+T^4-3T^3-2T^2+1)$	2, 2

Table 2: Elliptic-value polynomials at all fifteen prime levels, in the normalization $T_N = q^{-1}+O(q)$, from Jorgenson et al. (2019, §4.2, Table 1). The indices follow the displayed factors; each index applies to every root of its factor. At admissible primes, the supersingular roots of $\overline{\mathfrak{h}}_N$ are exactly the thick values removed in (1.4).

Proof. For a level-one form f of weight w, $f|_w W_N = N^{w/2} f(N\tau)$. The form $E_{p-1}(N\tau)$ has q-expansion $\equiv 1 \equiv E_{p-1}(\tau)$, so it equals A modulo p by Fact I1(i); and $N^{(p-1)/2} \equiv \left(\frac{N}{p}\right)$. The other two identities hold in characteristic 0: $\Delta(-1/(N\tau))\Delta(-1/\tau) = N^{12}\tau^{24}\Delta(N\tau)\Delta(\tau)$, and T is w_N-invariant. □

Proposition 3.4 (weighted supersingular polynomial)**.** *Let $S(T) = \prod_{j=1}^{s}(T - t_j)^{(p+1)/e_j}$. Then $S \in \mathbb{F}_p[T]$ and*

$$S(\overline{T}) = \Delta_N^{-k} = q^{-(N+1)k} \prod_{n\geq 1} (1-q^n)^{-24k}(1-q^{Nn})^{-24k} \quad \text{in } \mathbb{F}_p((q)).$$

Equivalently, $S = A^{p+1}/\Delta_N^k$ as functions on $X_0(N)^+_{\mathbb{F}_p}$. In particular $\sum_j e_j^{-1} = (N+1)(p-1)/24$.

Proof. The function $\Sigma := A^{p+1}/\Delta_N^k$ is a ratio of mod-p forms of weight p^2-1. By Lemma 3.3 it is w_N-invariant, since $\left(\frac{N}{p}\right)^{p+1} = 1$. Δ_N does not vanish on $Y_0(N)$, so Σ is regular off the cusp of $X_0(N)^+$. Its q-expansion there is $\Delta_N^{-k} = q^{-(N+1)k}(1+O(q))$, because A has q-expansion 1. Hence Σ is a monic polynomial in T of degree $(N+1)k$. By Fact I1(ii) and Lemma 3.1(ii), Σ vanishes exactly at the supersingular t_j, to order $(p+1)/e_j$ in $T - t_j$. Comparing degrees gives the mass identity. □

Corollary 3.5. *The degree of the reduced denominator is*

$$\deg D_{\text{thin}} = \frac{(N+1)(p-1)}{24} - \sum_{e_i>1} \frac{1}{e_i}.$$

In particular $R_{N,p} = 0$ exactly when no thin supersingular value exists.

Proof. Separate the thin and non-thin terms in the mass identity of Proposition 3.4; use Theorem 1.2(b) for the last assertion. □

4 The Fricke residue formula

We now apply the arithmetic identities to the Fricke setting of Section 1. The derivative argument gives the pole support and local residues; Theorem 1.1 determines the signed weighted product.

4.1 The derivative and pole support

The following is the level-N version of Kaneko and Zagier (1998, §10, (35)–(38)), who credit the method to Koike.

Lemma 4.1. *In $\mathbb{F}_p((q))$, with $d/dT = (\theta T)^{-1}\theta$,*

$$\frac{d\overline{\psi}}{dT} = T^{p-1} - (\theta T)^{p-1}, \qquad (\theta T)^{p-1} = g(\overline{T}), \quad g := \frac{(\theta T)^{p-1}}{A^2} \in \mathbb{F}_p(T).$$

Proof. Over Λ_N, $\theta\psi = T^{p-1}\theta T - (\theta T)(q^p)$, and $(\theta T)(q^p) \equiv (\theta T)^p$. Divide by $\theta T = -q^{-1} + \cdots$, which is invertible. The function g is a ratio of mod-p forms of weight $2(p-1)$, it is w_N-invariant by Lemma 3.3, and it has q-expansion $(\theta T)^{p-1}$ because $A \equiv 1$. □

Lemma 4.2 (local structure of g)**.** *Let $t \in \mathbb{F}_{p^2}$ be a non-cuspidal point with effective inertia e_t, and put $\varepsilon_t = 1$ if t is supersingular and 0 otherwise. Then*

$$\operatorname{ord}_{T=t} g = \frac{(p-1)(e_t-1) - 2\varepsilon_t}{e_t}.$$

For thin t and x over t, let $a = A/(\theta T)^{(p-1)/2}$, a function on $X_0(N)$ near x. Then a has a simple zero at x,

$$a'(t) := \frac{da}{dT}(x) = \frac{(B/12)(x)}{(\theta T)(x)^{(p+1)/2}} \neq 0, \qquad g = a'(t)^{-2}(T-t)^{-2}(1+O(T-t)).$$

Proof. By Fact I1(iv) and Lemma 3.1(ii), θT vanishes to order $e_t - 1$ in a stack parameter z, and A to order ε_t; also $T - t \sim z^{e_t}$. This gives the order formula. For thin t, the map $X_0(N) \to X_0(N)^+$ is étale at x and $\theta T(x) \neq 0$. In characteristic 0, for forms f and M of equal weight w, $\theta(f/M) = (\partial f \cdot M - f \cdot \partial M)/M^2$, where $\partial = \theta - \frac{w}{12}E_2$ is the Serre derivative. Apply this with $f = E_{p-1}$ and $M = (\theta T)^{(p-1)/2}$ and reduce modulo p; at x we have $A(x) = 0$. The form ∂E_{p-1} has weight $p+1$ and q-expansion $\equiv \theta(1) + \frac{1}{12}E_2 \equiv \frac{1}{12}E_{p+1}$, using the Kummer congruence $2(p+1)/B_{p+1} \equiv 24$ and $\sigma_p \equiv \sigma_1$. So $\partial A = B/12$ as mod-p forms. Finally $da/dT = \theta a/\theta T$. The value $B(x) \neq 0$ because $B/12$ is the derivative of A at its simple zero. □

Proof of Theorem 1.2(b)–(c). Let $t \in \mathbb{F}_{p^2}$. By Proposition 2.4, near t we have $\overline{\psi} = -\overline{H}(t^p, t)(T-t)^{-1} + O(1)$, because $T^{p^2} - T = -(T-t)(1+O(T-t))$. So the coefficient of $(T-t)^{-2}$ in $d\overline{\psi}/dT$ is $\overline{H}(t^p, t)$. By Lemma 4.1 it is also $-[g]_{-2}$. Hence

$$\operatorname{Res}_t R_{N,p} = -\overline{H}(t^p, t) = [g]_{-2}.$$

By Lemma 4.2, $\operatorname{ord}_t g \geq 0$ unless t is thin. For $p \geq 5$ and $e_t \geq 2$ we have $(p-1)(e_t - 1) - 2 \geq 0$. At thin t, $[g]_{-2} = a'(t)^{-2} = 144(\theta T)(x)^{p+1}/B(x)^2 \neq 0$. Every pole is simple by Proposition 2.4, so $R_{N,p} = P/D_{\text{thin}}$ with $P(t_i) \neq 0$ at every thin t_i. Since $R_{N,p}(\infty) = 0$, $\deg P < \deg D_{\text{thin}}$. □

4.2 The weighted signed product

Proposition 4.3 (weighted residue product)**.** *For thin t_i and $x = (E, C)$ over t_i,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \beta(E)\, D_N(x)^k.$$

Proof. Near x, Proposition 3.4 gives $S(T) = A^{p+1}/\Delta_N^k = a^{p+1}\,\xi^k$ with $\xi := (\theta T)^{12}/\Delta_N$, a function that is regular and nonzero at x. Hence $W_i = [S/(T-t_i)^{p+1}]_{T=t_i} = a'(t_i)^{p+1}\xi(x)^k$. With $r_i = a'(t_i)^{-2}$ this gives

$$r_i W_i = a'(t_i)^{p-1}\xi(x)^k = \Big(\frac{B}{12}\Big)^{p-1}(\theta T)^{-\frac{p^2-1}{2}} \cdot \frac{(\theta T)^{\frac{p^2-1}{2}}}{\Delta_N^k} = \frac{B^{p-1}}{\Delta_N^k}(x).$$

Since $\Delta_N = \Delta(\tau)\Delta(N\tau) = \Delta^2/D_N$, we have $\Delta_N^k = \Delta^{2k} D_N^{-k}$, which gives the second equality. □

Proof of Theorem 1.2(e). Proposition 4.3 and Theorem 1.1 give directly

$$r_i W_i = \beta(E)\, D_N(x)^k = -\left(\frac{-1}{p}\right)\left(\frac{N}{p}\right) = -\left(\frac{-N}{p}\right).$$

By Corollary 2.6, $r_i \in \mathbb{F}_p^\times$, and hence $W_i = -\left(\frac{-N}{p}\right) r_i^{-1} \in \mathbb{F}_p^\times$. □

Proof of Theorem 1.2(f). Proposition 2.4 gives $\overline{H}(t^p, t) = -\operatorname{Res}_t R_{N,p}$. By part (b) this is zero unless $t = t_i$ is thin, in which case it equals $-r_i = \left(\frac{-N}{p}\right) W_i^{-1}$ by part (e). □

Remark 4.4 ($\mathbb{F}_p$-rationality at non-rational points). The two rationality assertions in Theorem 1.2(e) have different explanations. The assertion $r_i \in \mathbb{F}_p$ follows from symmetry of the Hecke polynomial and Frobenius-equivariance of residues (Corollary 2.6). In contrast, writing

$$W_i = \operatorname{Nm}_{\mathbb{F}_{p^2}/\mathbb{F}_p}(D'_{\text{thin}}(t_i)) \prod_{e_j > 1} (t_i - t_j)^{(p+1)/e_j},$$

shows that $W_i \in \mathbb{F}_p$ requires the product of the thick-point factors to lie in $\mathbb{F}_p$. This is not visible factor by factor; it follows here from the signed product identity together with Corollary 2.6: $W_i = -\left(\frac{-N}{p}\right)/r_i \in \mathbb{F}_p^\times$.

5 Geometric interpretation of the weights

The proof of Theorem 1.2 is algebraic. We now interpret its weights in the integral modular curve, retaining the geometric explanation of pole cancellation. Nothing in Sections 2–4 depends on this section, and the argument given for Proposition 5.1 is an outline that rests on the cited results of Deligne–Rapoport and Liu. Let $\mathbb{W}$ be the ring of Witt vectors of $\overline{\mathbb{F}}_p$ and let

$$\mathcal{B} = X_0(Np)^{\mathrm{DR}}_{\mathbb{W}}, \qquad \mathcal{A} = \mathcal{B}/\langle w_N \rangle.$$

Here $\mathcal{B}$ denotes the coarse Deligne–Rapoport model and w_N is the partial Atkin–Lehner involution. A split node with completed local ring $\mathbb{W}[[s,t]]/(st - p^e v)$, v a unit, has *thickness* e.

Proposition 5.1. *The special fibre of $\mathcal{A}$ has two components, each isomorphic to $X_0(N)^+_{\overline{\mathbb{F}}_p}$. They meet at the supersingular Fricke orbits. At the orbit of $x = (E, C_N)$ the node has thickness*

$$e_x = |\operatorname{Aut}(E, C_N)/\{\pm 1\}|\,|\operatorname{Stab}_{\langle w_N \rangle}(x)|.$$

Consequently $R_{N,p}$ has a pole at a supersingular value precisely when the corresponding node has thickness one.

Outline of proof. Deligne–Rapoport's theorem applies with prime-to-p level N and the stabilizer of a line in $(\mathbb{Z}/N)^2$ as the auxiliary group (Deligne and Rapoport, 1973, V, 1.14–1.20; VI, Theorem 6.9, especially (6.9.1)). It gives two copies of $X_0(N)_{\overline{\mathbb{F}}_p}$ with Frobenius gluing, and the coarse completed local ring at x is

$$\mathbb{W}[[s,t]]/(st - p^{m_x}), \qquad m_x = |\operatorname{Aut}(E, C_N)/\{\pm 1\}|.$$

The forgetful map $\mathcal{B} \to X_0(N)_{\mathbb{W}}$ is finite: it is proper and has finite fibres on the generic fibre and on the two special-fibre components. Normality identifies $\mathcal{B}$ with the normalization in its generic function field. The generic partial Fricke map covers w_N on $X_0(N)_{\mathbb{W}}$, hence extends uniquely by normalization. On the elliptic locus it is

$$(E, C_N, C_p) \longmapsto (E/C_N, E[N]/C_N, q_N(C_p)).$$

Since q_N is prime to p, it preserves the Frobenius and Verschiebung components and both branches at a node.

To apply Liu's quotient result, note that $\mathcal{B}$ is a normal, proper flat fibred surface over the DVR $\mathbb{W}$, with smooth generic fibre and nodal special fibre. At a fixed geometric node its stabilizer has order two, prime to p, and acts trivially on its residue field $\overline{\mathbb{F}}_p$. As just shown it preserves both branches. It acts nontrivially on either coarse branch. Indeed its action on the smooth curve $X_0(N)$ is a nontrivial tame involution, whose derivative at a fixed point is -1. Taking the branch-preserving quotient therefore multiplies the thickness by this order, by Liu (2002, Proposition 10.3.48(b)–(c), pp. 526–528). At a non-fixed node the inertia is trivial and the thickness is unchanged. Invariants commute with reduction because 2 is invertible in $\mathbb{W}$. The quotient components are consequently $X_0(N)^+_{\overline{\mathbb{F}}_p}$, and no node is smoothed, since no stabilizer exchanges branches. This proves the thickness formula. The last assertion is Theorem 1.2(b). □

Remark 5.2 (level two). For $N = 2$, the Fricke fixed points are $u = \pm 64$, with $T = -104, 152$. At $u = -64$ the reduced automorphism group has order two and the Fricke stabilizer has order two, giving thickness four. At $u = 64$ their orders are one and two, giving thickness two. All other supersingular values have thickness one. The two exceptional values are supersingular precisely when $\left(\frac{-1}{p}\right) = -1$ and $\left(\frac{-2}{p}\right) = -1$, respectively. Thus a thick value is absent from the pole divisor but remains in the weighted residue product. The same distinction applies at all fifteen levels.

6 Finite certificates and computational evidence

For Theorem 1.2, exact-arithmetic computations cover all fifteen levels in $\mathcal{N}$ for $5 \le p \le 47$, excluding $p = N$. They reconstruct $R_{N,p}$ independently of the signed residue formula and then verify that formula. They are checks on the theorems, not inputs to their proofs. The following bound specifies the strength of the certificates.

Proposition 6.1 (finite certificate)**.** *Let $L = p^2$, let $D \in \mathbb{F}_p[T]$ be a monic divisor of $T^L - T$, of degree d, and let $P \in \mathbb{F}_p[T]$ have degree less than d. If*

$$D(\overline{T})\,\overline{T|U_p} - P(\overline{T}) = O(q^{L-d+1}),$$

then $R_{N,p} = P/D$. For $D = 1$, take $P = 0$.

Proof. Proposition 2.4 implies that $R_{N,p} - P/D = C/(T^L - T)$ with $\deg C < L$. If $C \neq 0$, its q-order is $L - \deg C$, since $T = q^{-1} + O(q)$. Multiplication by $D(T)$ gives order $L - \deg C - d \le L - d$, contradicting the hypothesis. Thus $C = 0$. □

For each pair, the supersingular values and effective indices are determined from the modular model, giving D_{thin}. A separate q-expansion calculation recovers the numerator without using the residue signs and checks Proposition 6.1, using source precision at least $p^3 + 2$. The reduced fraction, mass identity, weighted supersingular polynomial and signed residue products are also checked.

At $N \in \mathcal{N}_0$, the inputs are the eta coordinate and its rational j-map. At the other ten levels, we use the standard smooth elliptic or hyperelliptic models $v^2 = f_N(u)$ with $w_N(u, v) = (u, -v)$ and relations

$$j + j \circ w_N = \alpha_N(u), \qquad j(j \circ w_N) = Q_N(u),$$
$$j = \frac{\alpha_N(u) + \beta_N(u)v}{2}, \quad Q_N = \frac{\alpha_N^2 - f_N\beta_N^2}{4}.$$

These are the model data used by SageMath's prime-isogeny routines (SageMath Developers, 2026). The coordinate is $T_N = u - s_N$, where $s_N = -[u^{N-1}]\alpha_N/N$. Formal Newton inversion of $\alpha_N(u(q)) = j(q) + j(q^N)$ gives its expansion, with the product relation checked separately. Supersingular fibres satisfy $j_0^2 - j_0\alpha_N(u) + Q_N(u) = 0$; the roots of f_N detect Fricke ramification, and fibre multiplicities over $j_0 = 0, 1728$ give the reduced automorphism orders. Thus the indices are obtained independently of both the signed residue formula and the weighted Hasse polynomial.

All 184 admissible pairs pass the finite certificate. They contain 4458 thin values, of which 3448 lie outside $\mathbb{F}_p$ (Table 3). At every thin value the weighted residue product is $-\left(\frac{-N}{p}\right)$; pairs with no thin values have an empty residue-product set.

Further tests were run at the five eta-coordinate levels $N \in \mathcal{N}_0$. All 222 admissible pairs through $p = 211$ are certified (6246 thin values, 4696 outside $\mathbb{F}_p$); recovery of the denominator without a candidate was checked through $p = 37$, and Hecke polynomials modulo p^2 through $p = 31$. A further 960 pairs for $223 \le p \le 1499$ were checked using the double-pole formula of Lemma 4.2; these are not independent reconstructions from $T|U_p$.

Remark 6.2 (size of the input data). Corollary 1.3 reconstructs $R_{N,p}$ from the supersingular values and their effective indices. By the mass identity of Proposition 3.4 and Lemma 3.2, there are at most $(N+1)(p-1)/24 + \deg \mathfrak{h}_N$ such values. By contrast, the certificate of Proposition 6.1 uses the coefficients of $T|U_p$ up to q^{p^2-d}, where $d = \deg D_{\text{thin}}$, that is, those of T up to $q^{p(p^2-d)}$. This compares the size of the input only. Obtaining the supersingular values has its own cost, and no claim about running times is made here; timings for particular implementations are recorded in the supplementary material.

N	certified pairs	thin values	outside $\mathbb{F}_p$
2	13	33	8
3	13	47	16
5	12	67	32
7	12	92	54
11	12	136	86
13	12	163	116
17	12	203	140
19	12	231	170
23	12	267	202
29	12	334	260
31	12	350	272
41	12	445	352
47	12	483	382
59	13	733	614
71	13	874	744
Total	184	4458	3448

Table 3: Certified reconstructions at all fifteen levels for $5 \leq p \leq 47$, $p \neq N$. Values are distinct geometric T_N-values within each pair (N, p); counts are summed over the tested primes.

Data and software. The supplementary material for this paper consists of five packages. Their folder names are those of the archived release; they do not reflect the current status or numbering of the results.

- `Signed_Weighted_Residue_v1_0`: code, per-pair JSON results and reproduction instructions for the residue formula at $N \in \mathcal{N}_0$ (`python-flint` 0.9.0 and `mpmath`).
- `Fricke_Fifteen_Levels_Extension_v1_0`: the verifier for the ten levels outside $\mathcal{N}_0$, a regression run at the five levels in $\mathcal{N}_0$, the model coefficients with their provenance, and per-pair certificates (`python-flint` 0.9.0).
- `Fricke_Section8_Unified_Certificates_v1_0`: the records and reproduction script for Table 3.
- `Fricke_Elliptic_Values_Fifteen_Levels_v1_0`: the discriminants and model calculations for Lemma 3.2.
- `Corollary_1_2_Reconstruction_Benchmark_v1_0`: timings for the reconstruction of Corollary 1.3 (numbered Corollary 1.2 in an earlier draft) at $N \in \mathcal{N}_0$, with the timing code, exact inputs and outputs, raw timing samples and verification records.

The code, data and reproduction instructions are publicly available at https://github.com/bbrhuft/fricke-signed-residues/tree/main/supplementary.

7 Further directions

The discriminant-ratio identity of Theorem 1.1(ii) holds without a genus-zero hypothesis, including for arbitrary prime-to-p isogenies (Jordan, 2026). The residue formula additionally uses an integral genus-zero quotient coordinate, the Hasse divisor and the Kodaira–Spencer derivative. Lemma 2.1 supplies this coordinate uniformly for all fifteen prime Fricke levels; a coordinate on $X_0(N)$ is unnecessary. At composite levels, the cusp structure and the chosen Atkin–Lehner quotient require separate analysis. For a quotient of positive genus, there is no global Hauptmodul, and the residue reconstruction must be reformulated geometrically. These extensions of the residue theorem remain research questions; neither is established here.

Code and data availability

All code, scripts and logs are available in the GitHub repository

https://github.com/bbrhuft/fricke-signed-residues

The computational materials described here are those of release v2.0 of this repository. Earlier versions of this work, in which this paper and its companion formed a single manuscript, are archived on Zenodo at doi:10.5281/zenodo.24024359 (v1.11) and doi:10.5281/zenodo.24084789 (v1.13).

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